3 Written Questions

1. (18 pts) For each algorithm, determine the worst-, average-, and best-case asymptotic run times (1 pt each). In addition, describe what kind of vectors (i.e. sorted, reversed, or something else) achieve these bounds (1 pt each).

Algorithm	Worst-case	Average-case	Best-case
Bubble sort	$O(n^2)$ reverse sorted	$O(n^2)$ random	$O(n)$ sorted list
Insertion sort	$O(n^2)$ reverse sorted	$O(n^2)$ random	$O(n)$ sorted list
Selection sort	$O(n^2)$ Ascending evens -> descending odds	$O(n^2)$ random	$O(n^2)$ sorted list
Quicksort	$O(n^2)$ reverse sorted	$O(n \log n)$ random	$O(n \log n)$ sorted list with median pivot

3. (7 pts) For each of the graphs generated in (2), what do you see? Do the curves follow the patterns that we expect? Do certain algorithms seem faster than others in different cases? Are there any surprises? Be specific and refer back to your answers to question 3.1.

The curves do follow the patterns we expect, or stated above. Certain algorithms perform better or worse depending on the test case. For instance, Selection sort is much slower than the rest in the best case and quick sort is much faster than the rest in average cases. All perform equally poorly in worst case scenarios. Something that surprised me is that quick sort isn't the fastest in best case scenarios, which makes sense though as nlog(n) > n.